

PART 1 - POISSON GOODNESS-OF-FIT: CALCULATIONS

Hypothesis Test Setup

H0: The number of patient arrivals per fixed interval follows a Poisson distribution with mean $\lambda = N/n$.
H1: The number of patient arrivals per fixed interval does NOT follow a Poisson distribution.
Significance level, α 0.05

Table 1: Raw Interval Data and Lambda Calculation

Interval No.	Interval No.	Observed Arrivals (x)
1	80-<98	6
2	2	1
3	116-<134	3
4	134-<152	5
5	152-<170	1
6	170-<188	4
	Total (N)	20

n = number of intervals	6	
N = total arrivals	20	
lambda = N / n	3.333333333	

Table 2: Poisson Probabilities, Expected Frequencies, and Chi-Square Contributions (each x value taken individually)

x (arrivals)	(# intervals with this count)	POISSON(x,lambda,FALSE)	n*P(x)	(O-E)^2/E
0	0	0.0357	0.2140	0.2140
1	2	0.1189	0.7135	2.3198
2	0	0.1982	1.1891	1.1891
3	1	0.2202	1.3213	0.0781
4	1	0.1835	1.1010	0.0093
5	1	0.1223	0.7340	0.0964
6	1	0.0680	0.4078	0.8600
7+	0	0.0532	0.3192	0.3192
Total	6	1.0000	6.0000	5.0859

Degrees of Freedom and Critical Value

k = number of bins	8
p = parameters (lambda)	1
df = k - 1 - p	6
alpha	0.05
Critical value	12.5916
Calculated Chi-Square	5.0859

Decision rule Reject H0 if χ^2 calculated > χ^2 critical

Conclusion

Decision Fail to Reject H0

PART 2 - EXPONENTIAL GOODNESS-OF-FIT: CALCULATIONS

Hypothesis Test Setup

H0: The interarrival times follow an Exponential distribution with rate $\lambda = 1 / \text{mean}$.

H1: The interarrival times do NOT follow an Exponential distribution.

Significance level, α 0.05

Table 1: Given Interarrival-Time Frequency Data, Mean, and Lambda

Class	Lower (a)	Upper (b)	Midpoint	Observed O
0-<5	0	5	2.5	9
5-<10	5	10	7.5	7
10+*	10	15	12.5	3
Total (N)				19

val time = SUMPRODUCT	5.9211
lambda = 1 / mean	0.1689

Table 2: CDF at Boundaries, Interval Probability [F(upper) - F(lower)], Expected Frequency, Chi-Square

Class	Lower a	Upper b	lambda*a)	lambda*b)	P = F(b)-F(a)	Observed O	Expected E=N*P	(O-E)^2/E
0-<5	0	5	0	0.5702	0.5702	9	10.8339	0.3104
5-<10	5	10	0.5702	0.8153	0.2451	7	4.6564	1.1796
10+ (tail)	10	inf	0.8153	1	0.1847	3	3.5098	0.0740
Total					1.0000	19	19.0000	1.5641

Degrees of Freedom and Critical Value

k = number of classes	3
Parameters estimated (lambda)	1
df = k - 1 - p	1
alpha	0.05
Critical value	3.8415
Calculated	1.5641

Decision rule Reject H0 if χ^2 calculated > χ^2 critical

Conclusion

Decision **Fail to Reject H0**